



Ensemble Models & Random Forest Regression

A practical guide to ensemble methods for numerical predictive modeling using the `daltoolbox` package in R. Random Forest combines multiple decision trees to reduce variance and improve prediction accuracy — each example illustrates how ensemble techniques outperform single-model approaches while keeping the experimental workflow consistent throughout.

ENSEMBLE MODELS

RANDOM FOREST

R / DALTOOLBOX

BOSTON DATASET

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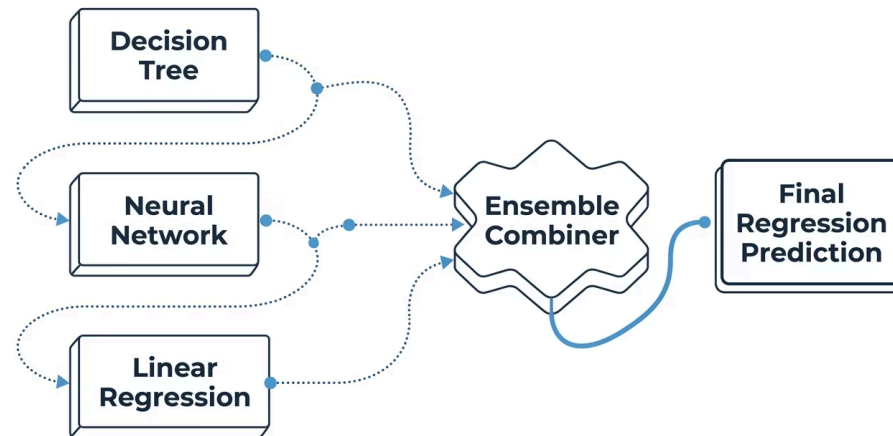
<https://eic.cefet-rj.br/~eogasawara>

Introduction to Ensemble Models for Regression

Ensemble methods in regression combine predictions from multiple individual models to achieve superior predictive performance compared to any single model.

This approach effectively reduces variance, improves accuracy, and captures complex non-linear relationships in data more robustly.

We will explore key techniques such as Random Forest, Gradient Boosting, and various model averaging strategies.



Ensemble models leverage the strengths of diverse predictors to output a more reliable and accurate forecast.

Random Forest: Theoretical Foundations

Random Forest is an ensemble method that combines multiple decision trees trained independently. Each tree is built on a bootstrap sample of the data and uses a random subset of variables at each split, making the model robust and low-variance.



Bagging (Bootstrap Aggregating)

Each tree is trained on a random sample *with replacement* from the training set. This ensures diversity among trees and reduces overfitting.



Random Variable Selection

At each split node, only a random subset of `mtry` variables is considered. This decorrelates the trees and further increases the diversity of the ensemble.



Variance Reduction

Averaging the predictions of many decorrelated trees reduces the variance of the final model without increasing bias, improving generalization.



Final Prediction

For regression, the final prediction is the **arithmetic mean** of the predictions from all `ntree` trees in the forest.

Main Hyperparameters

`mtry`

Number of variables randomly selected to evaluate at each node split. Smaller values increase diversity among trees; larger values bring the model closer to a single tree. Default recommendation: $p/3$ for regression.

`ntree`

Total number of trees in the forest. More trees increase prediction stability, but with growing computational cost. Typical values: 100 to 500 trees.

References: Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32. | Liaw, A. & Wiener, M. (2002). Classification and Regression by randomForest. *R News*, 2(3), 18–22.

reg_rf: Random Forest for Regression

Setup & Data

```
# Regression Random Forest

library(daltoolbox)
library(MASS)

data(Boston)
Boston <- as.matrix(Boston)

set.seed(1)
sr <- sample_random()
sr <- train_test(sr, Boston)
boston_train <- sr$train
boston_test <- sr$test
```

Training & Evaluation

```
# mtry: variables per split
# ntree: number of trees
model <- reg_rf("medv", mtry=7, ntree=30)
model <- fit(model, boston_train)

# Training evaluation
train_pred <- predict(model, boston_train)
train_eval <- evaluate(model,
  boston_train[, "medv"], train_pred)
print(train_eval$metrics)
# mse=1.36, smape=0.04, R2=0.98

# Test evaluation
test_pred <- predict(model, boston_test)
test_eval <- evaluate(model,
  boston_test[, "medv"], test_pred)
print(test_eval$metrics)
# mse=14.60, smape=0.12, R2=0.76
```

Step-by-Step

- Define the model with `reg_rf("medv", mtry=7, ntree=30)`, where `medv` is the target variable
- `mtry`: number of variables considered at each split
- `ntree`: number of trees in the ensemble
- Training $R^2=0.98$ indicates a near-perfect fit — watch for overfitting
- Test $R^2=0.76$ is the best generalization result achieved so far

❏ **Key Concept:** Random Forest averages many trees trained with randomness, effectively **reducing variance**. The gap between training and test performance (0.98 vs. 0.76) is expected and typical of ensemble methods. Ref: Breiman (2001).

Summary: Ensemble Models with daltoolbox

Random Forest as a Bagging Ensemble

The `reg_rf` combines multiple decision trees trained with bootstrap samples and random variable selection (`mtry`), reducing variance compared to a single tree. The final result is the average of predictions from all trees in the ensemble.

Results on the Boston Dataset

The model `reg_rf("medv", mtry=7, ntree=30)` achieved $R^2=0.98$ on training and $R^2=0.76$ on test — the best generalization result among the evaluated models. The gap between training and test performance is expected and typical of ensemble methods, reflecting variance reduction rather than severe overfitting.

The Only Regression Ensemble in daltoolbox

The package offers `reg_rf` as the only ensemble model for regression. Boosting and XGBoost are available only for classification (`cla_boosting`, `cla_xgboost`). For regression tasks, Random Forest remains the only native ensemble option in daltoolbox.

References & Additional Resources

Academic references and online resources to explore daltoolbox, Random Forest, and Ensemble Models in greater depth.

Online Resources



CRAN Package – daltoolbox

cran.r-project.org/web/packages/daltoolbox

DAL Toolbox – Documentation

cefet-rj-dal.github.io/daltoolbox/

GitHub Repository – daltoolbox

github.com/cefet-rj-dal/daltoolbox

Source code, issues, and contributions.

Key Academic References

- Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32.
- Liaw, A. & Wiener, M. (2002). Classification and Regression by randomForest. *R News*, 2(3), 18–22.
- Harrison, D. & Rubinfeld, D.L. (1978). Hedonic prices and the demand for clean air. *Journal of Environmental Economics & Management*, 5, 81–102. [Boston Dataset]
- Ogasawara, E. et al. (2026). daltoolbox: Leveraging Experiment Lines to Data Analytics. R package version 1.3.767. *CRAN*.
- Ogasawara, E. et al. (2009). Experiment Line: Software Reuse in Scientific Workflows. *SSDBM 2009, LNCS 5566*, pp. 264–272. *Springer*.